Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 1 of 13 · Question 2 part b.ii · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
2	(a)	• two bones; • coming together/meet;		2
	(b) (i)	X = tibia; Y = femur;	reject tibula	2
	(ii)	tendons • connects bone to muscle; • transmits force/pull;		2
		cartilage • reduces friction/shock absorber/cushions; • allows smooth movement;		2
	(iii)	• ligament; • second /another muscle;		2

Total 10 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 1 of 13 · Question 2 part b.ii · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 2 of 13 · Question 1 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

1	(a)	(i)	X cartilage; Y femur; Z knee cap;		1 1 1
		(ii)	**B** ligaments hold the leg bones together A is incorrect as ligaments do not attach muscles to bones C is incorrect as ligaments do not move leg bones D is incorrect as ligaments do not stop the leg bones rubbing together		1
		(iii)	hinge/synovial (joint)		1
	(b)		In the following order only oxygen; carbon dioxide;		 1 1
	(c)		biceps contracted/fatter; triceps relaxed/thinner; 		1 1

Total question 1 = 9 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 2 of 13 · Question 1 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 3 of 13 · Question 2 part b.iv · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.iv in this mark scheme extract

2(a)	A; (1 and 3) (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(i)	D spongy bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	B compact bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(iii)	cartilage;		**(1)**

Question Number	Answer	Additional guidance	**Mark**
2(b)(iv)	shock absorber/cushions; allows free movement of joint/prevents grinding/reduces friction;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)	• Ca^{2+}needed; • forms part of bone structure / provides strength to bone/no calcium bones are less dense/weaker; • protein required; • for muscle development/component • of muscle; • vitamin D needed; • to allow uptake of Ca^{2+};		**(6)**

Total 12 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 3 of 13 · Question 2 part b.iv · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.iv in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 4 of 13 · Question 3 part a.ii · 2 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
3	(a)	(i)	A = synovial fluid; B = synovial membrane;		2
		(ii)	• lubricate joint; • for easier movement/to reduce friction;		2
	(b)	(i)	• one section covering ball; • one section covering semi-circular part of bone; •		2
		(ii)	• movement not smooth/bones rub together/friction/less shock absorbing; • painful;		2
	(c)		• tendons attach to muscles; • attached to bone/lower arm; • muscle/biceps/triceps contracts; • generates pull on tendons/lower arm/bone; • tendons inelastic;	attaches muscles to bones = 2	5

Total : 13 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 4 of 13 · Question 3 part a.ii · 2 marks

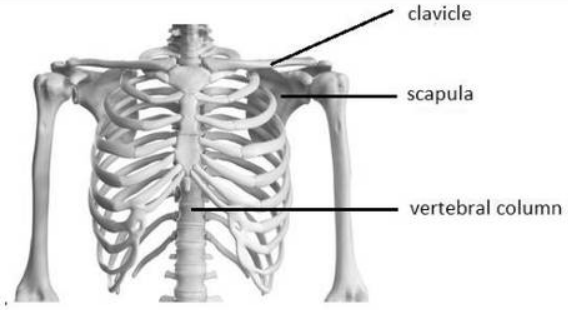
2022 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 5 of 13 · Question 1 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
1	(a)	(i)			3
		(ii)	• support the body; • protection (of the brain and vital organs);		2
	(b)		Three from: • decrease in bone mass/bones less dense/more spongy/porous; • bone strength weakens/brittle bones/bones break more easily/fractures occur; • height decreases/stoop/poor posture; • pain;		3

Total 8 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 5 of 13 · Question 1 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 6 of 13 · Question 2 part b.iii · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
2	(a)	• two bones; • coming together/meet;		2
	(b) (i)	X = tibia; Y = femur;	reject tibula	2
	(ii)	tendons • connects bone to muscle; • transmits force/pull;		2
		cartilage • reduces friction/shock absorber/cushions; • allows smooth movement;		2
	(iii)	• ligament; • second /another muscle;		2

Total 10 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 6 of 13 · Question 2 part b.iii · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 7 of 13 · Question 5 part b.i · 1 mark

2024 JUN · Paper 01 (regional R) · find part b.i in this mark scheme extract

5 (a)	(i)	Joint A = ball and socket; Joint B = hinge joint;		(2)
	(ii)	• joint A has movement with 3 degrees of freedom/3 planes of movement/ rotational ; • joint B can only move in one plane/ backwards and forwards;	Allow description	(2)
(b)	(i)	(ligament) holds bones together;		(1)
	(ii)	cartilage) reduces friction between bones / prevents bones rubbing together/ shock absorber;		(1)

Total for Question 5 = 6 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 8 of 13 · Question 5 part b.ii · 1 mark

2024 JUN · Paper 01 (regional R) · find part b.ii in this mark scheme extract

5 (a)	(i)	Joint A = ball and socket; Joint B = hinge joint;		(2)
	(ii)	• joint A has movement with 3 degrees of freedom/3 planes of movement/ rotational ; • joint B can only move in one plane/ backwards and forwards;	Allow description	(2)
(b)	(i)	(ligament) holds bones together;		(1)
	(ii)	cartilage) reduces friction between bones / prevents bones rubbing together/ shock absorber;		(1)

Total for Question 5 = 6 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 9 of 13 · Question 1 part a.ii · 3 marks

2025 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

1 (a) (i) long (bone) — 1

(ii)

Part	Function: reduces friction	Function: produces red blood cells	Function: provides strength
compact bone			✓
spongy bone		✓	
cartilage	✓		

3

(iii)

- provides nutrients/named nutrient/oxygen;
- removes waste/named waste

2

(b) (i) In the following order:

- tendons;
- contracts;
- relaxes;
- pulls;

4

(ii) 0.5 x 125;
62.5 (J);

Accept 63 (J)

2

Total for Question 1 = 12 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 10 of 13 · Question 8 part a.ii · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

8(a)(i)	broken down in liver; by deamination; to form urea; used to make new protein/amino acids;		**(3)**

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	poor/restricted movement of synovial joints pain in joint; backflow of blood in heart; poor/inefficient blood circulation/oxygenation; less energy/tired;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
8(b)(i)	• not sex-linked; • more females affected than males/more males would be affected; • only females would be carriers (if sex-linked); • male 1/3/11 would show condition if sex-linked; • but these males are carriers;		**(4)**

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	• probability that child has condition is ¼ /1:3 / 1 in 4/25%/0.25; • probability that child is male is ½/1 in 2/1:1 / 50%/0.5; • so probability that it is affected male is 12.5%/0.125/$\frac{1}{8}$;		**(3)**

Total 13 Marks

Paper Total 90 Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 10 of 13 · Question 8 part a.ii · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 11 of 13 · Question 4 part e · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

4 (a)	$\frac{30\,000}{100}$ x 40; 12 000;	Full marks for final correct answer ecf	1 1
(b)	a microorganism/named microorganism that causes disease;		1
(c)	It cannot be transmitted from one person to the next		1
(d)	• use a sample of people/use two groups of people; • feed each sample/group of people a different diet/feed one group a diet with vitamin B and the other without vitamin B; • monitor health/compare (health) of two groups;		1 1 1
(e)	• breakdown of cartilage/cartilage not formed properly/damaged; • bones rub together/more friction between bones/less shock absorption;		1 1

Total question 4 = 9 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 11 of 13 · Question 4 part e · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 12 of 13 · Question 6 part c · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

Question number				
6 (a)		• where two bones; • come together/meet;		2
(b)	(i)	• humerus; • ulna;		2
	(ii)	• covers end (of ball) of humerus/L; • covers depression in ulna/M;		2
(c)		• movement not smooth/bones grind/rub together/less shock absorption/increased friction; • painful;		2
(d)		any four from • attached to end of muscle/attach muscle to bone; • inelastic/do not stretch; • muscle contracts; • pulls tendon; • causing M/ulna/bone to move;		4 Total 12

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 12 of 13 · Question 6 part c · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 13 of 13 · Question 3 part b.ii · 2 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
3	(a) (i)	A = synovial fluid; B = synovial membrane;		2
	(ii)	• lubricate joint; • for easier movement/to reduce friction;		2
	(b) (i)	• one section covering ball; • one section covering semi-circular part of bone; •		2
	(ii)	• movement not smooth/bones rub together/friction/less shock absorbing; • painful;		2
	(c)	• tendons attach to muscles; • attached to bone/lower arm; • muscle/biceps/triceps contracts; • generates pull on tendons/lower arm/bone; • tendons inelastic;	attaches muscles to bones = 2	5

Total : 13 marks

Type MS · State the function of a skeletal or joint part and explain effects of damage

Answer 13 of 13 · Question 3 part b.ii · 2 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks